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ENGINEERING



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CO3-AT2 Data Interpretation

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1. Continuous Distribution — Area Interpretation

A continuous random variable X is normally distributed with $\mu = 100$, $\sigma = 15$.

(a) What proportion of values lie between 85 and 130? (b) A value of 145 is observed — how many standard deviations away is it, and would you flag it as a potential outlier under the empirical rule?

A continuous random variable X is normally distributed with:

Mean, $\mu = 100$

Standard deviation, $\sigma = 15$

We need to determine:

(a) The proportion of values between 85 and 130.

(b) How many standard deviations away 145 is and whether it should be flagged as a potential outlier.

(a) Proportion of values between 85 and 130

Since X follows a normal distribution, we first convert the given values into z-scores.

The formula for the z-score is:

$$z = (X - \mu) / \sigma$$

Where:

X = observed value

μ = population mean

σ = population standard deviation

Step 1: Calculate the z-score for 85

Given:

$$X = 85$$

$$\mu = 100$$

$$\sigma = 15$$

Therefore:

$$z_1 = (85 - 100) / 15$$

$$z_1 = -15 / 15$$

$$z_1 = -1$$

Therefore, 85 is 1 standard deviation below the mean.

Step 2: Calculate the z-score for 130

Given:

$$X = 130$$

$$\mu = 100$$

$$\sigma = 15$$

Therefore:

$$z_2 = (130 - 100) / 15$$

$$z_2 = 30 / 15$$

$$z_2 = 2$$

Therefore, 130 is 2 standard deviations above the mean.

Step 3: Find the area between the two z-scores

We now need:

$$P(85 < X < 130)$$

which is equivalent to:

$$P(-1 < Z < 2)$$

From the standard normal distribution table:

$$P(Z < 2) = 0.9772$$

$$P(Z < -1) = 0.1587$$

Therefore:

$$P(-1 < Z < 2) = P(Z < 2) - P(Z < -1)$$

$$= 0.9772 - 0.1587$$

$$= 0.8185$$

Step 4: Convert the proportion into a percentage

$$\text{Percentage} = 0.8185 \times 100$$

$$= 81.85\%$$

Final Answer for (a)

The proportion of values lying between 85 and 130 is:

0.8185

Therefore, approximately 81.85% of the observations are expected to lie between 85 and 130.

(b) A value of 145 is observed

We need to determine how many standard deviations 145 is away from the mean.

Using the z-score formula:

$$z = (X - \mu) / \sigma$$

Substituting:

$$z = (145 - 100) / 15$$

$$z = 45 / 15$$

$$z = 3$$

Therefore:

$$z = 3$$

This means that 145 is 3 standard deviations above the mean.

Interpretation Using the Empirical Rule

For a normal distribution, the empirical rule states:

- Approximately 68% of observations lie within 1 standard deviation of the mean.
- Approximately 95% of observations lie within 2 standard deviations of the mean.
- Approximately 99.7% of observations lie within 3 standard deviations of the mean.

The value 145 is exactly 3 standard deviations above the mean.

Therefore, it lies at the extreme end of the normal distribution.

2. Sampling Bias vs Sampling Error

A survey estimating average household expenditure samples only online panel respondents. The sample mean is ₹18,000 vs a known population mean of ₹14,500 (from census data). (a) Is this discrepancy better explained by sampling error or sampling bias? (b) Would increasing the sample size fix this problem? Justify.

A survey is conducted to estimate average household expenditure.

Given:

Sample mean = ₹18,000

Known population mean = ₹14,500

The survey samples only people from an online panel.

We need to determine whether the difference is caused by sampling error or sampling bias and whether increasing the sample size would solve the problem.

(a) Sampling Error or Sampling Bias?

First, calculate the difference between the sample mean and the population mean.

Difference:

Sample Mean – Population Mean

= ₹18,000 – ₹14,500

= ₹3,500

Therefore, the sample estimate is ₹3,500 higher than the known population average.

The percentage difference is:

Percentage Difference = $(3,500 / 14,500) \times 100$

$\approx 24.14\%$

Thus, the sample mean is approximately 24.14% higher than the population mean.

Sampling Error

Sampling error occurs because a sample rather than the entire population is studied.

Even when a sample is selected randomly, the sample mean may not exactly equal the population mean.

For example, if different random samples are selected from the same population, their sample means may differ slightly.

Sampling error is therefore a natural consequence of sampling.

Sampling Bias

Sampling bias occurs when the sampling method systematically produces a sample that is not representative of the population.

In this problem, only online panel respondents are surveyed.

This is important because households that participate in online panels may differ from households that are not represented in the panel.

For example, online panel respondents may differ in:

- Internet access
- Technology usage
- Age
- Income
- Education
- Employment
- Lifestyle
- Online purchasing behavior

Therefore, the sample may systematically overrepresent particular types of households.

Why Sampling Bias Is the Better Explanation

The problem does not simply say that a random sample happened to produce ₹18,000.

Instead, it specifically states that the sample contains only online panel respondents.

This is a problem with the **method of selecting the sample**.

Therefore, the difference is better explained by **sampling bias** rather than ordinary sampling error.

Answer for (a)

The discrepancy is primarily explained by:

Sampling bias

because the sample is restricted to online panel respondents and may not represent the entire population of households.

(b) Would Increasing the Sample Size Fix the Problem?

No.

Increasing the sample size primarily helps reduce **sampling error**, but it does not automatically eliminate **sampling bias**.

For example, suppose the original sample contains 100 online respondents.

Increasing it to 1,000 online respondents may make the sample mean more precise for the online-panel population.

However, if online-panel respondents are systematically different from the entire population, the larger sample may still produce a biased estimate.

The problem can be represented as:

Biased sampling method + Larger sample = More precise but potentially still biased estimate

Example

Suppose the true population mean is:

₹14,500

If the online panel consistently contains households with higher expenditure, a sample might produce:

₹18,000

Even if the sample is increased from 100 to 10,000 respondents, the estimate may remain higher than ₹14,500 if the selection process remains biased.

How to Fix the Problem

The researchers should improve the sampling method.

Possible approaches include:

- Random sampling
- Stratified random sampling
- Systematic sampling
- Probability sampling
- Proper representation of different household groups

For example, if households are divided into appropriate groups based on relevant characteristics and sampled proportionally, the resulting sample can better represent the population.

3. Covariance Sign vs Correlation Strength

Two variable pairs both show positive covariance: Pair 1 has Cov = 850, Pair 2 has Cov = 12. Explain why you cannot conclude Pair 1 has a "stronger relationship" than Pair 2 without additional information, and state what that missing information isTwo variable pairs have positive covariance.

Pair 1:

$$\text{Cov}(X,Y) = 850$$

Pair 2:

$$\text{Cov}(X,Y) = 12$$

We need to explain why Pair 1 cannot automatically be considered to have a stronger relationship than Pair 2.

Step 1: Understand What Covariance Indicates

Covariance describes the direction in which two variables tend to move together.

If covariance is positive:

As one variable increases, the other variable tends to increase.

If covariance is negative:

As one variable increases, the other variable tends to decrease.

Therefore:

Pair 1:

$$\text{Cov}(X,Y) = +850$$

This indicates a positive relationship.

Pair 2:

$$\text{Cov}(X,Y) = +12$$

This also indicates a positive relationship.

Step 2: Why the Values 850 and 12 Cannot Be Compared Directly

It may seem that:

$$850 > 12$$

Therefore:

Pair 1 must have a stronger relationship.

However, this conclusion is incorrect.

The reason is that covariance depends on the units and scales of measurement of the variables.

For example, suppose one variable is measured in rupees and another in kilograms.

If the rupee values are converted to smaller or larger units, the numerical value of covariance changes.

Therefore, covariance does not have a standardized scale.

Step 3: What Is Missing?

To compare the strength of the relationships, we need:

- Standard deviation of X
- Standard deviation of Y

The Pearson correlation coefficient is calculated as:

$$r = \text{Cov}(X,Y) / (\sigma_x \times \sigma_y)$$

Where:

$\text{Cov}(X,Y)$ = covariance

σ_x = standard deviation of X

σ_y = standard deviation of Y

Step 4: Why Correlation Is Better for Comparing Strength

Correlation is standardized and always lies between:

-1 and +1

The interpretation is:

r close to +1 → strong positive relationship

r close to 0 → weak or no linear relationship

r close to -1 → strong negative relationship

Example

Suppose Pair 1 has:

$$\text{Cov}(X,Y) = 850$$

but:

$$\sigma_x = 100$$

$$\sigma_y = 100$$

Then:

$$r = 850 / (100 \times 100)$$

$$r = 850 / 10,000$$

$$r = 0.085$$

This would indicate a weak positive linear relationship.

Now suppose Pair 2 has:

$$\text{Cov}(X,Y) = 12$$

$$\sigma_x = 4$$

$$\sigma_y = 4$$

Then:

$$r = 12 / (4 \times 4)$$

$$r = 12 / 16$$

$$r = 0.75$$

This represents a much stronger positive relationship.

Therefore, the smaller covariance can actually correspond to the stronger relationship.

4. Estimation — Confidence Interval Reasoning

A sample of $n = 100$ items has mean = 250 and SD = 40. Construct a 95% confidence interval for the population mean. Then explain, in interpretation (not formula) terms, what this interval does and does NOT tell you about the true population mean.

Given:

Sample size:

$$n = 100$$

Sample mean:

$$\bar{x} = 250$$

Sample standard deviation:

$$s = 40$$

Confidence level:

$$95\%$$

We need to construct a 95% confidence interval for the population mean.

Step 1: Determine the Standard Error

The standard error of the sample mean is:

$$SE = s / \sqrt{n}$$

Substitute the values:

$$SE = 40 / \sqrt{100}$$

Since:

$$\sqrt{100} = 10$$

Therefore:

$$SE = 40 / 10$$

$$SE = 4$$

Thus:

$$\text{Standard Error} = 4$$

Step 2: Determine the Critical Value

For a 95% confidence interval using the large-sample normal approximation:

$$z = 1.96$$

This means that the central 95% of the standard normal distribution lies between approximately -1.96 and $+1.96$.

Step 3: Calculate the Margin of Error

The formula is:

$$\text{Margin of Error} = z \times \text{SE}$$

Substitute:

$$\begin{aligned}\text{Margin of Error} &= 1.96 \times 4 \\ &= 7.84\end{aligned}$$

Therefore:

$$\text{Margin of Error} = 7.84$$

Step 4: Calculate the Lower Confidence Limit

$$\begin{aligned}\text{Lower Limit} &= \text{Sample Mean} - \text{Margin of Error} \\ &= 250 - 7.84 \\ &= 242.16\end{aligned}$$

Step 5: Calculate the Upper Confidence Limit

$$\begin{aligned}\text{Upper Limit} &= \text{Sample Mean} + \text{Margin of Error} \\ &= 250 + 7.84 \\ &= 257.84\end{aligned}$$

Step 6: Write the Confidence Interval

Therefore:

$$\begin{aligned}95\% \text{ Confidence Interval} \\ &= 250 \pm 7.84 \\ &= (242.16, 257.84)\end{aligned}$$

Final Answer

$$95\% \text{ Confidence Interval} = (242.16, 257.84)$$

Therefore, the estimated population mean lies within the interval:

242.16 to 257.84

Interpretation of the Confidence Interval

The confidence interval gives a range of plausible values for the population mean based on the sample information.

The sample mean is 250.

The calculated interval is:

242.16 to 257.84

This means that the sample provides evidence consistent with population mean values in this range at the 95% confidence level.

What the Confidence Interval Tells Us

The interval tells us about the uncertainty involved in estimating a population mean from a sample.

If the same population were sampled repeatedly and a 95% confidence interval were calculated each time using the same method, approximately 95% of those intervals would contain the true population mean.

What the Confidence Interval Does Not Tell Us

It does not mean that there is a 95% probability that the fixed population mean is between 242.16 and 257.84.

The true population mean is a fixed but unknown quantity.

The 95% refers to the reliability of the procedure used to construct the interval.

It also does not mean that 95% of individual observations lie between 242.16 and 257.84.

The confidence interval concerns the **population mean**, not individual observations.

Final Interpretation

The sample estimate of the population mean is 250, and the 95% confidence interval extends from 242.16 to 257.84. The interval expresses the uncertainty associated with estimating the population mean from the sample.

5. Integrated Case Study

A retail company's daily sales data (in ₹'000) for 15 days is:

120, 125, 118, 130, 122, 128, 500, 119, 124, 121, 126, 123, 117, 129, 125

(a) Identify the outlier and justify using both the IQR method and z-score method.

(b) Compute mean and median with and without the outlier — compare the sensitivity of each.

(c) Compute skewness with the outlier included — classify the distribution shape.

(d) Recommend whether the company should remove, cap, or investigate the outlier before reporting monthly performance, and justify your recommendation using the statistical evidence from (a)–(c).

A retail company's daily sales data, measured in ₹'000, for 15 days is:

120, 125, 118, 130, 122, 128, 500, 119, 124, 121, 126, 123, 117, 129, 125

We need to:

(a) Identify the outlier using IQR and z-score.

(b) Calculate mean and median with and without the outlier.

(c) Calculate skewness with the outlier included.

(d) Recommend whether to remove, cap, or investigate the outlier.

(a) Identify the Outlier Using IQR and Z-Score

Step 1: Arrange the Data

The original data are:

120, 125, 118, 130, 122, 128, 500, 119, 124, 121, 126, 123, 117, 129, 125

Arrange them in ascending order:

117, 118, 119, 120, 121, 122, 123, 124, 125, 125, 126, 128, 129, 130, 500

Step 2: Calculate the Median

There are 15 observations.

For an odd number of observations:

Median position = $(n + 1) / 2$

= $(15 + 1) / 2$

= $16 / 2$

= 8

The 8th value is:

124

Therefore:

Median = 124 thousand

IQR Method

Step 3: Calculate Q1

Using the standard percentile calculation:

$$Q1 = 120.5$$

Step 4: Calculate Q3

$$Q3 = 127$$

Therefore:

$$IQR = Q3 - Q1$$

$$= 127 - 120.5$$

$$= 6.5$$

Thus:

$$\text{IQR} = 6.5 \text{ thousand}$$

Step 5: Calculate the Lower Outlier Boundary

The formula is:

$$\text{Lower Boundary} = Q1 - 1.5(IQR)$$

Substitute:

$$\text{Lower Boundary} = 120.5 - 1.5(6.5)$$

First calculate:

$$1.5 \times 6.5 = 9.75$$

Therefore:

$$\text{Lower Boundary} = 120.5 - 9.75$$

$$= 110.75$$

Step 6: Calculate the Upper Outlier Boundary

$$\text{Upper Boundary} = Q3 + 1.5(IQR)$$

$$= 127 + 1.5(6.5)$$

$$= 127 + 9.75$$

$$= 136.75$$

Therefore:

$$\text{Lower Boundary} = 110.75$$

$$\text{Upper Boundary} = 136.75$$

Step 7: Compare the Observations with the Boundaries

The normal range according to the IQR rule is:

$$110.75 \leq X \leq 136.75$$

Most observations lie between 117 and 130.

However:

$$500 > 136.75$$

Therefore:

500 is an outlier according to the IQR method.

The value 500 is measured in ₹'000.

Therefore, the outlier represents:

$$₹500 \times 1,000 = ₹500,000$$

Z-Score Method

Step 8: Calculate the Mean Including the Outlier

First calculate the sum of all observations:

$$\Sigma X = 120 + 125 + 118 + 130 + 122 + 128 + 500 + 119 + 124 + 121 + 126 + 123 + 117 + 129 + 125$$

$$\Sigma X = 2227$$

Number of observations:

$$n = 15$$

Therefore:

$$\text{Mean} = \Sigma X / n$$

$$= 2227 / 15$$

$$= 148.4667$$

Therefore:

$$\text{Mean} \approx 148.47 \text{ thousand}$$

Step 9: Calculate the Standard Deviation

Using the population standard deviation formula:

$$\sigma = \sqrt{[\Sigma(X - \mu)^2 / N]}$$

For this dataset:

$$\Sigma(X - \mu)^2 \approx 132,619.73$$

Therefore:

$$\sigma^2 = 132,619.73 / 15$$

$$\approx 8,841.32$$

Taking the square root:

$$\sigma \approx \sqrt{8,841.32}$$

$$\sigma \approx 94.03$$

Therefore:

Population standard deviation ≈ 94.03 thousand

Step 10: Calculate the Z-Score of 500

The z-score formula is:

$$z = (X - \mu) / \sigma$$

For $X = 500$:

$$z = (500 - 148.47) / 94.03$$

$$z = 351.53 / 94.03$$

$$z \approx 3.74$$

Therefore:

$$\mathbf{z \approx 3.74}$$

Step 11: Interpret the Z-Score

A commonly used rule is:

$$|Z| > 3 \rightarrow \text{Potential outlier}$$

Here:

$$|3.74| > 3$$

Therefore:

500 is a potential outlier according to the z-score method.

Conclusion for Part (a)

Both methods identify the same observation:

Outlier = 500 thousand

IQR method:

$$500 > 136.75$$

Z-score method:

$$3.74 > 3$$

Therefore, there is strong statistical evidence that the ₹500,000 sales observation is unusual compared with the other daily sales values.

(b) Mean and Median With and Without the Outlier

Mean With the Outlier

The total of all 15 observations is:

$$\Sigma X = 2227$$

$$n = 15$$

$$\text{Mean} = 2227 / 15$$

$$\text{Mean} \approx 148.47$$

Therefore:

Mean with outlier = ₹148.47 thousand

or:

₹148,466.67 approximately

Median With the Outlier

The ordered data are:

117, 118, 119, 120, 121, 122, 123, 124, 125, 125, 126, 128, 129, 130, 500

There are 15 observations.

Median position:

$$(15 + 1) / 2 = 8$$

$$8\text{th value} = 124$$

Therefore:

Median with outlier = ₹124 thousand

Remove the Outlier

Remove 500.

The remaining observations are:

117, 118, 119, 120, 121, 122, 123, 124, 125, 125, 126, 128, 129, 130

Mean Without the Outlier

Original sum:

$$2227$$

Outlier:

$$500$$

New sum:

$$2227 - 500 = 1727$$

Number of observations:

14

Therefore:

$$\text{Mean} = 1727 / 14$$

$$\text{Mean} = 123.3571$$

Therefore:

Mean without outlier $\approx$ ₹123.36 thousand

Median Without the Outlier

There are now 14 observations.

For an even number of observations, the median is the average of the 7th and 8th values.

The 7th value is:

123

The 8th value is:

124

Therefore:

$$\text{Median} = (123 + 124) / 2$$

$$= 247 / 2$$

$$= 123.5$$

Therefore:

Median without outlier = ₹123.50 thousand

Comparison of Mean and Median

Measure With Outlier Without Outlier Change

Mean	148.47	123.36	25.11
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Median	124.00	123.50	0.50
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Change in Mean

$$\text{Change} = 148.47 - 123.36$$

$$= 25.11 \text{ thousand}$$

Therefore, the mean decreases by approximately:

₹25.11 thousand

when the outlier is removed.

Percentage Change in Mean

Percentage change:

$$= (25.11 / 148.47) \times 100$$

$$\approx 16.91\%$$

Therefore, the outlier increases the reported mean by approximately 16.91%.

Change in Median

Change:

$$124 - 123.5$$

$$= 0.5 \text{ thousand}$$

Therefore, the median changes by only:

$$\text{₹}0.50 \text{ thousand}$$

This is very small compared with the change in the mean.

Interpretation

The results clearly demonstrate that the mean is highly sensitive to extreme observations.

The single value of 500 increases the mean from:

$$\text{₹}123.36 \text{ thousand}$$

to:

$$\text{₹}148.47 \text{ thousand}$$

However, the median changes only from:

$$\text{₹}123.50 \text{ thousand}$$

to:

$$\text{₹}124.00 \text{ thousand}$$

Therefore, the median provides a much more stable measure of typical daily sales when an extreme value is present.

(c) Calculate Skewness With the Outlier Included

Step 1: Understand Skewness

Skewness measures the degree and direction of asymmetry in a distribution.

The general interpretation is:

$$\text{Skewness} \approx 0 \rightarrow \text{approximately symmetric}$$

Positive skewness → longer right tail

Negative skewness → longer left tail

Because the dataset contains a very large value of 500 compared with the other observations, we expect the distribution to be positively skewed.

Step 2: Use the Moment Coefficient of Skewness

The population moment coefficient of skewness is:

$$\text{Skewness} = \mu_3 / \sigma^3$$

where:

$$\mu_3 = \Sigma(X - \mu)^3 / N$$

and:

σ = population standard deviation

From the dataset:

Mean:

$$\mu = 148.4667$$

Population standard deviation:

$$\sigma \approx 94.0283$$

The sum of cubed deviations is approximately:

$$\Sigma(X - \mu)^3 = 43,203,025.85$$

Therefore:

$$\mu_3 = 43,203,025.85 / 15$$

$$\mu_3 \approx 2,880,201.72$$

Now calculate:

$$\sigma^3 = 94.0283^3$$

$$\sigma^3 \approx 831,333.61$$

Therefore:

$$\text{Skewness} = 2,880,201.72 / 831,333.61$$

$$\text{Skewness} \approx 3.4646$$

Therefore:

$$\text{Skewness} \approx 3.46$$

Interpretation of Skewness

The calculated skewness is approximately:

+3.46

The positive sign indicates that the distribution is positively skewed, or right-skewed.

The magnitude of 3.46 is quite large, indicating strong positive skewness.

The reason for this is the observation:

500

Most of the observations are concentrated between approximately 117 and 130, while 500 lies far to the right of these values.

This creates a long right-hand tail.

Conclusion for Part (c)

The sales distribution with the outlier included is:

Strongly positively skewed (right-skewed).

The ₹500,000 observation is the primary reason for this strong positive skewness.

(d) Recommendation — Remove, Cap, or Investigate the Outlier?

The company has three possible choices:

1. Remove the outlier
2. Cap the outlier
3. Investigate the outlier

Option 1: Remove the Outlier

The company should not automatically remove the observation.

Although 500 is statistically identified as an outlier, it may represent a genuine business event.

For example, the transaction could be:

- A bulk customer purchase
- A large corporate order
- A seasonal sale
- A special promotion
- A major one-time contract
- A large institutional customer order

Removing a genuine transaction would result in loss of useful business information.

Option 2: Cap the Outlier

Capping means replacing an extreme value with a selected upper threshold.

For example, the company could replace 500 with a predetermined upper limit.

However, this should not be done automatically.

If 500 is a genuine sale, replacing it with a lower value would distort the actual sales record.

Therefore, capping should only be considered when there is a documented statistical or business reason for doing so.

Option 3: Investigate the Outlier

Investigation is the most appropriate first step.

The company should check:

- Was the transaction entered correctly?
- Was the quantity entered correctly?
- Was the price entered correctly?
- Was the transaction duplicated?
- Was there a bulk purchase?
- Was there a special promotion?
- Was it a seasonal event?
- Was it a genuine large customer order?

Statistical Evidence Supporting Investigation

There is strong evidence that the value is unusual.

Evidence 1: IQR Method

Upper IQR boundary:

136.75 thousand

Observed value:

500 thousand

Since:

$500 > 136.75$

the observation is an IQR outlier.

Evidence 2: Z-Score

Z-score:

3.74

Since:

$$3.74 > 3$$

the observation is a potential z-score outlier.

Evidence 3: Effect on Mean

Mean with outlier:

148.47 thousand

Mean without outlier:

123.36 thousand

Difference:

25.11 thousand

The mean changes by approximately:

16.91%

Evidence 4: Effect on Median

Median with outlier:

124 thousand

Median without outlier:

123.5 thousand

Difference:

0.5 thousand

Therefore, the median is relatively stable.

Evidence 5: Skewness

Skewness:

+3.46

This indicates strong positive skewness.

The extreme value of 500 is therefore substantially influencing the shape of the distribution.

Final Recommendation

The company should **investigate the outlier first rather than immediately removing or capping it.**

The statistical analysis clearly shows that ₹500,000 is an unusual observation:

- It exceeds the IQR upper boundary.
- Its z-score is approximately 3.74.
- It increases the mean by approximately 16.91%.

- It has very little effect on the median.
- It causes strong positive skewness of approximately 3.46.

However, statistical unusualness does not automatically mean that the observation is incorrect.

If the ₹500,000 transaction is genuine

The company should retain it in the actual sales records because it represents a real business event.

For reporting typical daily performance, the company should consider presenting the median along with the mean and separately identifying the exceptional transaction.

If the ₹500,000 transaction is an error

If investigation shows that the value resulted from a duplicate transaction, incorrect quantity, incorrect price, or data-entry mistake, the company should correct or remove the erroneous record before final reporting.

If the transaction is genuine but represents a one-time event

The company may retain the transaction but treat it separately during forecasting.

For example, if the company knows that ₹500,000 resulted from a one-time corporate order that is unlikely to occur again, blindly using it as normal demand could lead to overestimating future demand.

This could result in excessive inventory purchases.

Therefore, exceptional events should be identified separately when developing forecasts.